Supporting information for *The Selling closure: a complete operator set for reindexing, and the price of pseudo-symmetry*

Petrus H. Zwart *

Molecular Biophysics and Integrated Bioimaging Division, Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA.

E-mail: phzwart@lbl.gov

All code is in `agentsg` (Zwart, 2026), version 0.5.0, `github.com/phzwart/agentsg`; the scripts named below are in its directory `manuscript/benchmark`. Table, section and equation numbers of the main text are cited as there. Conorm vectors are written in the slot order da, db, dc, ab, ac, bc of the main text, and strains in per cent. Section S7 lists, for every section, table and figure, the check that reproduces its numbers.

S1. Catalogues of shapes and specialisations

Cell shapes. Table S1 lists the shapes of the ten sorts with more than one, each as the cell with the most right angles and no acute angle. The three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the extra shapes of primitive orthorhombic, tetragonal and cubic lattices are monoclinic-looking cells with two right angles; C-centred orthorhombic has two primitive shapes; the boundary sorts of body-centred orthorhombic, C-centred monoclinic and triclinic have two or three cells with a single right angle each, for triclinic the known instability of the reduced cell when an angle is near 90° (Andrews *et al.*, 1980).

Table S1. *The cell shapes of the sorts with more than one, for the cells named; each shape is given as the cell with the most right angles and all angles* $\geq 90°$*, shortest edges first when several qualify and b unique for the monoclinic-looking ones, with the number n of members (up to swaps) that have it. Centred cells are given by their primitive cells; the boundary sorts are specified by their conorms in* $Å^2$.

sort	lattice	n	shape
cP	30 Å	4	30, 30, 30; 90, 90, 90
		12	30, 30, 42.4; 90, 135, 90
tP	30, 30, 50 Å	4	30, 30, 50; 90, 90, 90
		4	30, 30, 58.3; 90, 121.0, 90
		8	30, 50, 42.4; 90, 135, 90
oP	30, 40, 55 Å	4	30, 40, 55; 90, 90, 90
		4	30, 40, 62.6; 90, 118.6, 90
		4	30, 55, 50; 90, 126.9, 90
		4	40, 55, 50; 90, 143.1, 90
oS	C 30, 40, 50 Å	2	25, 50, 25; 90, 106.3, 90
		4	25, 50, 30; 90, 126.9, 90
mP	30, 40, 50 Å, 100°	2	30, 40, 50; 90, 100, 90
		2	30, 40, 53.7; 90, 113.4, 90
		2	50, 40, 53.7; 90, 146.6, 90
oI (O3)	900, 1600, 0, 0, 1600, 900	1	50, 50, 50; 111.1, 129.8, 90
		2	42.4, 50, 50; 90, 115.1, 115.1
mC (M2B)	900, 1600, 2500, 0, 1600, 900	1	50, 50, 70.7; 104.7, 116.9, 90
		1	50, 50, 65.6; 105.9, 105.9, 90
mC (M3)	2500, 1600, 0, 0, 1600, 900	1	50, 50, 64.0; 120, 90, 111.1
		2	50, 56.6, 58.3; 90, 108.0, 124.4
aP (A2)	2500, 400, 1200, 0, 1600, 900	1	36.1, 60.8, 64.0; 114.3, 90, 114.2
		1	36.1, 56.6, 64.0; 116.2, 90, 101.3
aP (A3)	2500, 400, 0, 0, 1600, 900	1	36.1, 50, 53.9; 90, 101.9, 119.9
		1	44.7, 50, 53.9; 90, 99.6, 135.7
		1	36.1, 44.7, 58.3; 90, 115.3, 104.4

Specialisations. The conorms of any member, in any labelling, are integer combinations of the reduced cell's conorms, hence linear in the free parameters of the sort (the letters of Table S10). Imposing another sort's zeros and equalities on one member is a homogeneous linear system in those parameters, and a solution with non-negative parameters, a non-singular lattice and a resulting lattice of the target sort and no higher is a specialisation. Table S2 lists the conditions of codimension one, on 36 pairs of sorts. Each is a parameter vanishing or two parameters becoming equal. The classical values appear: body-centred tetragonal is face-centred cubic at $c/a = \sqrt{2}$ and body-centred cubic at $c/a = 1$; C-centred orthorhombic is hexagonal at $b = \sqrt{3}\,a$ and tetragonal at $b = a$; rhombohedral is cubic at $\alpha = 60°$, 90° and 109.47°. The mono-

clinic conditions come in threes, one per shape: with unique axis c and plane lattice a, b, the lattice is C-centred orthorhombic when $|a| = |b|$, $|a| = |a + b|$ or $|b| = |a + b|$, and primitive orthorhombic when $a \perp b$, $a \perp (a+b)$ or $b \perp (a+b)$ — the textbook condition $a = c$ read in each of the three cell choices. The table also shows which sorts are one condition from a much higher symmetry: C-centred monoclinic from tetragonal, body-centred tetragonal and rhombohedral; the body-centred orthorhombic boundary sort from face-centred cubic; triclinic on a wall from C-centred monoclinic.

Table S2. *All specialisations of codimension one between the 24 sorts. Conditions are on the letters of the sort's pattern in Table S10; "shapes" is the number of distinct cell shapes of the starting sort through which the conditions are reached, and the last column reads the condition in the cell of the conventional description where that is short.*

from	to	conditions	shapes	reading
aP A1	aP A2	any letter $= 0$	1	a right angle appears
aP A2	aP A3	$v = 0$	1	a second right angle
aP A2	mC M2A	$s = t$, $r = u$	1	
aP A2	mP M4	r, s, t or $u = 0$	1	
aP A3	mC M3	any two letters equal	3	one per shape
mP M4	oP O5	$r = 0$, $s = 0$, $t = 0$	3	$a \perp b$, $a \perp (a + b)$, $b \perp (a + b)$
mP M4	oS O4	$s = t$, $r = t$, $r = s$	3	$\lvert a\rvert = \lvert b\rvert$, $\lvert a\rvert = \lvert a+b\rvert$, $\lvert b\rvert = \lvert a+b\rvert$
mC M1A	oF O1A	$r = t$	1	
mC M1A	oI O2	$u = 0$	1	
mC M1A	oS O4	$r = 0$, $t = 0$	1	
mC M1B	mC M2B	$t = 0$, $u = 0$	1	boundary sort
mC M1B	mC M3	$r = 0$, $s = 0$	1	boundary sort
mC M1B	oF O1A	$r = s$	1	
mC M1B	oI O1B	$t = u$	1	
mC M2A	oI O2	$r = t$	1	
mC M2B	oI O3	$t = 0$	1	
mC M2B	tI T2	$r = s$	1	one step to tI
mC M2B	tP T5	$r = 0$, $s = 0$	1	one step to tP
mC M3	oI O3	$r = t$	1	
mC M3	hR R3	$r = s$, $s = t$	1	one step to hR
oP O5	tP T5	$r = s$, $r = t$, $s = t$	1	$a = b$, $a = c$, $b = c$
oS O4	tP T5	$r = 0$	1	$b = a$
oS O4	hP H4	$r = s$	1	$b = \sqrt{3}\,a$
oI O1B	oI O3	r, s or $t = 0$	1	boundary sort
oI O1B	tI T1	$r = s$, $r = t$, $s = t$	1	two edges equal
oI O2	tI T2	$r = s$	1	
oI O3	cF C3	$r = s$	1	one step to cF
oF O1A	tI T1	$s = t$	1	
oF O1A	tI T2	$s = 0$, $t = 0$	1	
tP T5	cP C5	$r = s$	1	$c = a$
tI T1	cI C1	$r = s$	1	$c/a = 1$
tI T1	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from below
tI T2	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from above
hR R1	cP C5	$r = 0$	1	$\alpha = 90^\circ$
hR R1	cI C1	$r = s$	1	$\alpha = 109.47^\circ$
hR R3	cF C3	$r = s$	1	$\alpha = 60^\circ$

S2. The benchmark, the comparisons and the walks

The benchmark. The directory `manuscript/benchmark` of the repository holds 780 measured cells with exact references and true operators, in five families: 75 cells measured in a different shape of the same lattice; 660 lattices placed 0.5 to 8 times the error from a condition of Table S2; 20 cells whose only reduced superbase is a difference-wall

member; 16 needles and plates with angular errors; 9 cells of the boundary sorts. Each case carries its Niggli cell. The errors are isotropic in the metric tensor, of median strain 0.1%. A case counts as found when the true proper operator itself is among the candidates. `evaluate_strain.py` scores the closure and the $\{-1, 0, 1\}$ search at the band and at three times each case's error (`results_strain.txt`); `evaluate.py --niggli` tallies the Niggli cells by family and sort (`results_niggli.txt`). The measured Niggli cell is a different cell from the reference's in 45 cases (O1B 23, O3 12, M1B 6, one each at M2B, M3, R3 and T2; 14 of them a change between type I and type II), and the same cell in another basis in 125 (O1B 25, M1A 15, O1A 15, T1 15, M1B 13, O4 9, O2 7, R1 7, M2A 6, T2 6, O3 5, A2 1, A3 1).

The comparisons. `compare_methods.py` runs the searches of Table S13 on the benchmark (`results_compare.txt`) and, with `--stress`, on the stress family (`results_stress.txt`): 60 references drawn from the benchmark, each with independent traceless errors on the reference and the new cell, scaled so that the true operator's price is t times the band, $t = 0.5$, 0.9, 1.1 and 1.5, and a random proper reindexing with entries in $\{-2, \ldots, 2\}$. Each method's candidates are made proper, priced by the geodesic distance after the best scale, and kept within the band; at $t = 1.1$ and 1.5 no method found the true operator, as by construction none should. `window_failures.py` builds the window family (`results_window_failures.txt`): 300 lattices of the sorts M4, O4, H4, O5, T5 and C5 with random parameters, each measured twice with an error of 0.1 to 2%, the new cell in a random proper basis and the reference given as its Niggli cell, classified by the Niggli type of the reference and by the kind of face (with an exact reference in the labelling of the sort's pattern the zeros never sit on d and no window fails); the same 300 lattices, drawn again, with the reference in a random proper basis; and 60 lattices on random two- and three-zero faces whose closure has a member needing an entry of 3, measured in

that member. `coset_rank.py` groups the closure's candidates by the exact symmetries of the underlying lattice (`results_coset.txt`). `niggli_types.py` reduces near-monoclinic cells from their Niggli cells and records where the near-zero conorms land (`results_niggli_types.txt`). `face_labellings.py` gives, for every face, the largest entry of the operators out of the reduced cell as reduction returns it and in its best labelling (`results_face_labellings.txt`). `lookup_benchmark.py` runs the lookup of $\mathcal{U}$ on the benchmark and the stress family and counts the references whose r_1 does not certify it (`results_lookup.txt`). `benchmark_misses.py` lists the benchmark cases that the $\{-1, 0, 1\}$ window and the pipeline miss (`results_benchmark_misses.txt`). `niggli_presentation.py` reduces measured near-oP cells from their Niggli form and from random bases and records where the near-zero conorms land (`results_niggli_presentation.txt`). `lookup_plate_case.py` builds the plate case in which an uncertified lookup misses an operator (`results_lookup_plate_case.txt`). `export_universal_operators.py` writes $\mathcal{U}$ as column operators with their determinants (`universal_operators_columns.json`). `adjacency_radii.py` computes r_1 for the benchmark references and for 2000 PDB cells drawn at random, and times the widening and the reindexing (`results_adjacency_radii.txt`).

The PDB. `pdb_nonisomorphism.py` measures the price between entries of one crystal form (`results_pdb_nonisomorphism.txt`). `pdb_lepage.py` runs the cctbx lattice-symmetry analysis at 3° on every entry and prices the pseudo-symmetries (`pdb_lepage.jsonl`, `results_pdb_lepage.txt`); `pdb_pricing_summary.py` gives the transitions and the fractions within the non-isomorphism band (`results_pdb_pricing.txt`). `pdb_band_controls.py` varies the cut between crystal forms, drops identical cells, weights deposition blocks equally, splits the band by Bravais type, and counts on 3000 same-form pairs how often a wrong class is cheaper than the identity's (`results_pdb_band_controls.txt`). `pdb_lepage_null.py`

compares the Le Page rate of real cells with that of cells whose parameters are shuffled within a space group (`results_pdb_lepage_null.txt`). Table S8 lists the prices by the Delone sort of the Le Page lattice and Table S9 by transition from the reported type.

Walks. Table S3 compares four walks on the measured metric, with a band τ on the conorms, and the construction of Section S4 against brute force.

Table S3. *Four walks on the measured metric, and the construction of Section S4, against brute force on 80 cells with zero conorms at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns), with a band τ on the conorms at* 0.1 *and* 0.5 *of* $\lambda_1/32$.

procedure	cells with a wrong set
flip wherever $\lvert p\rvert \le \tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le \tau$, filter at the end	13
flip wherever $\lvert p\rvert \le 2\tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

S3. The operators alone

The closure can be described without any lattice. An operator $\mathbf{W}$, written as the new basis in the coordinates of the reduced cell, acts on the six conorms by a fixed integer 6×6 matrix $\mathbf{A_W}$, since every conorm of $\mathbf{W}F_0$ is an integer combination of the conorms of F_0. The reduced cells are the points of the orthant $\{p \ge 0\}$ in conorm space. Selling (1874) showed that every lattice has a conorm vector in this orthant, unique up to the 48 swaps except on the faces (Delaunay, 1933; Conway & Sloane, 1992), and the closure of a lattice is the set of points of its orbit that lie in the orthant: one in the interior, several on the faces. So $\mathbf{W}$ is an operator of the closure of every lattice on the face of Z exactly when $\mathbf{A_W}$ maps the face $\{p \ge 0,\ p_Z = 0\}$ into the orthant, a sign condition on the entries of $\mathbf{A_W}$ and nothing else.

Enumerating integer matrices with entries in $[-2, 2]$ under that condition, over all 38 faces, and closing the result under relabelling at both ends, gives a set $\mathcal{U}$ of

$6096 = 48\times127$ integer matrices with entries in $\{-3,\dots,3\}$; per face there are 48×1, 2, 3, 6 and 16 matrices for no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector and three zeros. Both placements of three zeros, the triangle and the path, were enumerated; they describe the same lattices read from an orthogonal cell or from a cell with a face diagonal, each has 16 members, and no swap carries one placement into the other. $\mathcal{U}$ is closed under inversion, and every element has a labelling whose basis vectors have entries in $[-2,2]$. That $\mathcal{U}$ contains every operator between two reduced cells of one lattice, whatever the lattice, does not rest on that window: Lemma A2 of the main text confines the members of every closure to superbase coordinates $s+2k$ with $|k_i|\le 3$ relative to a reduced cell, and the enumeration of that whole window, over all 38 faces, finds only members that lie in $\mathcal{U}$ in each of their 48 labellings. In the basis of the reduced cell their entries are in $\{-1,0,1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 4 of the main text). Up to relabelling at both ends $\mathcal{U}$ has nine kinds of element, of flip length 0 to 4 (Table S4).

Table S4. *The nine kinds of operator between Selling-reduced cells, up to relabelling at both ends; "matrices" counts the integer matrices of the kind in $\mathcal{U}$, "flips" its length as a flip word, and the last column the zero patterns in whose closures it occurs.*

kind	matrices	flips	occurs with
1	48	0	every pattern (the swaps)
2	288	1	every pattern with a zero
3	1152	2	two zeros sharing a vector; three zeros
4	1152	3	three zeros
5	1152	3	three zeros
6	576	3	two zeros sharing a vector; three zeros
7	576	4	three zeros
8	576	4	three zeros
9	576	4	three zeros

Why four. For a lattice with $a \perp b \perp c$ the sixteen members are four orthogonal superbases, differing by the sign of one axis, and twelve with a face diagonal. A flip from an orthogonal cell always gives a cell with a face diagonal, so changing the sign of one axis takes three flips, out, across and in, and the diagonal cells attached only to the far orthogonal cell take one more: from (a,b,c) the members lie at flip distances $0,1,2,3,4$ in numbers $1,3,6,3,3$, and four is the diameter of the largest face.

S4. A band on the conorms

The simplest widening of the closure is an absolute band τ in Å^2 on the conorms,

$$\mathcal{S}_\tau(\mathbf{G}) = \{\, F \;:\; \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \,\}, \tag{1}$$

Selling reduction with its boundary moved from 0 to $-\tau$. This is an L^∞ band on the conorms, a different norm from the strain of the main text; the two are comparable only for cells whose edges are similar in length. It can be computed by idealising: set every conorm of the reduced superbase F_0 that is at most 4τ to zero, enumerate the exact closure of the idealised lattice Λ^*, and keep the members whose conorms under the measured metric are all at least $-\tau$. The construction is exact when λ_1, the squared length of the shortest lattice vector, exceeds 32τ (Theorem S4.5), a ratio independent of the scale of the cell: for an isotropic cell it allows an angular error of about 1.8°. It fails for anisotropic cells, where the band must cover the conorms between the long vectors: on the 16 needles and plates of the benchmark it declined 8. On lattices of five zero patterns with zero conorms placed at random it was exact at every band from $\lambda_1/32$ to $\lambda_1/6$ in which the idealised lattice existed; on dense two-zero grids idealising at 2τ was already exact, at τ not. For the worked cell at $\tau = 20\,\text{Å}^2$, within the hypothesis, it returns the six members of the ring. A new cell constructed so that its only reduced superbase is a difference-wall member of an orthorhombic corner was matched against two stored sets of the reference: the complete set offers three candidates and their negatives; a set built by idealising only at the band has lost the third, $\pm\mathbf{I}$, the true operator.

Bands per slot. A band that scales with the vectors of each superbase, $p_{ij}(F) \geq -\sigma|u_i||u_j|$ on every slot of F (all pairwise angles at least $90° - \arcsin\sigma$), is the natural angular form of the L^∞ band, but the same construction applied with each slot of F_0 idealised against its own band is not exact. On the needle $5.58 \times 5.21 \times 131.97\,\text{Å}$,

$\gamma = 91.12^\circ$, at $\lambda_1/(\sigma l_a l_c) = 33$ and 40, it returns 6 superbases where brute force finds 12: the conorm ab survives idealisation because its own band is small, while the members built on $a + c$ carry the same dot product on a slot whose band is about 25 times larger. Idealising until the zeroed slots are consistent across the members of the closure repairs this, and on 240 random needles, plates, isotropic and monoclinic cells it is never wrong (exact in 124, refused in 116), while the single-pass construction is wrong in 9 (z57).

For a superbase F and two metrics, $\kappa_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition S4.1 (characterisation superbase by superbase). $\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{C}(\mathbf{G}')$ for some $\mathbf{G}'$ with $\kappa_F(\mathbf{G}, \mathbf{G}') \le \tau\}$. *Proof.* ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\kappa_F \le \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) The metric whose conorms in the frame of F are $\max(p_{ij}, 0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite and F is obtuse for it. □

Lemma S4.2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class, the excess taken for Λ^*. *Proof.* In the frame of F_0, Lemma A1 of the main text gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, F is obtuse for Λ^*; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$. □

Lemma S4.3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. Lemma A1 in the frame of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau \sum_N \gamma$. If a large γ were positive, Lemma S4.4(ii)

would give $8\tau \sum_{\text{large}} \gamma < \tau \sum_N \gamma \le 8\tau \sum_{\text{large}} \gamma$; so the large γ vanish, and Lemma S4.4(i) gives $e_c \le 4\tau \cdot 4$. □

Lemma S4.4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) If $\sum_L \gamma_{ij} > 0$, then $\sum_N \gamma_{ij} \le 8 \sum_L \gamma_{ij}$. Both bounds are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, some $|k_i| \ge 16$, and since L connects the four vertices with $k_0 = 0$, $Q = \sum_L \Delta k^2 \ge 16^2/3 > 75$. For each pattern (L, N) let $\rho_{L,N}$ be the largest generalised eigenvalue of the quadratic form $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$; the largest over all pairs (L, N) is $3 + 2\sqrt{2}$. Then $\sum_N \gamma \le \rho_{L,N} Q + \sqrt{|N| \rho_{L,N} Q}$ and $\sum_L \gamma \ge Q - \sqrt{|L| Q}$, and the ratio of these bounds, $(\rho_{L,N}\sqrt{Q} + \sqrt{|N|\rho_{L,N}})/(\sqrt{Q} - \sqrt{|L|})$, decreases in Q, so it suffices to check it at $Q = 75$ for each pattern with its own $\rho_{L,N}$, where it is below 8 in every case; with the uniform value $3 + 2\sqrt{2}$ it would not be. The lemma is computer-assisted. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$. □

Theorem S4.5. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{C}(\Lambda^*) : p_{ij}(F; \mathbf{G}) \ge -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma S4.3 gives $e_c(F; \mathbf{G}) \le 16\tau$; the excess for Λ^* is the same sum with the zeroed slots dropped, so $e_c(F; \Lambda^*) \le 16\tau < 4m^*$, and Lemma S4.2 makes F obtuse for Λ^*. □

The operators from F_0 to the members of $\mathcal{C}(\Lambda^*)$ are integer matrices independent of the metric, so one enumeration serves every band up to the one idealised at.

Lemma S4.6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member

of the closure for Z' is a member of the closure for Z. *Proof.* The members for Z' are reached from F_0 by words of flips, each at a slot that is zero at that point of the word (Lemma A3 of the main text); a flip at a slot zero for both lattices acts on both conorm vectors as the same transposition of two slots, so after each flip the superbase is obtuse for both and the next flip is again at a slot zero for both. □

Proposition S4.7 (reindexing guarantee in the reference frame). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference's reduced frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \le \xi < \nu_{\min}$ with $\nu_{\min}$ the smallest eigenvalue of $\mathbf{G}_R$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \ge \sqrt{2}\,\xi\,V/(\nu_{\min} - \xi)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* With U the coordinates of F_N's basis vectors in the reference frame and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$, $\sum |u_i|^2 \le V/(\nu_{\min} - \xi)$; in F_N's own frame the error has Frobenius norm at most $\sum |u_i|^2\,\xi$, and every conorm shift is at most $\sqrt{2}$ times that norm. The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem S4.5 applies. □ For fifteen references near zero conorms of five zero patterns, each with twelve new cells carrying a random metric error and a random reindexing, the true operator was among the candidates in 180 of 180 cases at the band given by the coverage bound below; the guarantee of Proposition S4.7 is conservative, a median 13ξ against a band near 3ξ that was enough.

From a metric error to a band. A metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$, at most $c_{ij}\xi$ with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F$; every member survives every error of norm ξ exactly when $\tau \ge \tau_{\text{cov}} = \max(c_{ij}\xi - p_{ij})$, read off the stored closure. The largest constant on zero conorms is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and $2\sqrt{3}$ in six of the sixteen labelled cuboid frames, so $\tau = 3\xi$ is not a worst-case guarantee in every frame. These frame-dependent constants are what the

strain norm removes.

S5. Further tables

Table S5. *The seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024): the program's Bravais type and its minimal distance to that type (Å²), and the same divided by the mean squared edge (%); the Bravais types of the sort spectrum in increasing strain distance (%), aP omitted, up to and including the program's type; and the next type with its distance. For all seven cells r_1 reaches the cap, so "> 25" is a lower bound on the distance as well as the outcome of the minimum over the frames of $\mathcal{U}$. A band of* 7.4% *selects the program's type in all seven; the largest ratio between consecutive distances follows it in six, and for the alumina cell lies between mC and hR.*

sample	type	Å²	%	types up to the program's (strain %)	next
HERMES_Sr327_250K	tI	0.57	1.24	mC 0.31, oI 0.74, oF 0.83, tI 0.84	mP > 25
1339prbasrcuo_bs1	tP	0.056	0.10	mP 0.11, mC 0.14, oS 0.17, oP 0.18, tP 0.22	oF > 25
sample3	cI	2.07	0.78	mC 0.23, hR 0.39, oF 0.39, oI 0.49, tI 0.49, cI 0.58	cF > 25
CeO2_matsushita	cF	0.084	0.61	mC 0.09, oI 0.17, oF 0.17, tI 0.18, hR 0.26, cF 0.30	cI > 25
M402-pf-119673	hP	20.5	5.27	mC 0.99, mP 2.24, oS 2.37, hP 2.94	oI > 25
Acrinol-B-D8CuKa1	mC	0.72	0.36	mC 0.31	oF 20.6
Alumina676	hR	1.93	8.10	mC 0.43, hR 3.58	oI 14.9

Table S6. *The widened closure of the measured worked cell. Each row is a member, named by a word in the flips of Table 1 of the main text from the reduced cell e (the empty word); the six numbers are its conorms under the measured metric, in Å²; "depth" is the strain at which it enters, in %. At a strain of* 7.7%*, the widening for the default band, all sixteen are members.*

word	*da*	*db*	*dc*	*ab*	*ac*	*bc*	depth
e	49.5	1588.9	2444.7	4.2	847.5	4.5	0
bc	45.0	8.7	2449.3	1593.4	852.0	−4.5	0.28
ab	851.7	1593.1	2440.6	−4.2	53.7	8.7	0.49
bc.db	2458.0	−8.7	53.7	1602.1	843.3	4.2	0.62
ab.bc	842.9	4.5	2449.3	1601.8	62.5	−8.7	0.78
ab.bc.db	2453.8	−4.5	847.5	1606.3	57.9	−4.2	0.78
da	−49.5	2494.3	1638.4	53.7	897.0	−45.0	3.76
ab.bc.ac	1664.2	−57.9	2511.7	905.4	−62.5	53.7	4.56
da.ab	950.7	2548.0	1584.7	−53.7	4.2	8.7	5.06
da.ab.ac	−49.5	2543.8	1588.9	954.9	−4.2	12.9	5.08
ab.ac	49.5	1539.3	2494.3	905.4	−53.7	62.5	5.08
da.ab.bc	942.0	−45.0	1593.4	2556.7	12.9	−8.7	5.09
bc.db.dc	45.0	2511.7	−53.7	1548.4	897.0	57.9	5.09
ab.ac.bc.ac	1610.5	959.1	2565.5	−4.2	−8.7	−53.7	5.11
da.ab.bc.ac	2569.6	−57.9	1606.3	954.9	−12.9	4.2	5.11
ab.ac.bc	−12.9	967.8	2556.7	1601.8	8.7	−62.5	5.11

Table S7. *The sort patterns of Andrews & Bernstein (2023), in the S^6 slot order bc, ac, ab, ad, bd, cd, and the same patterns in the slot order da, db, dc, ab, ac, bc of Table S10. The representative used here is their pattern read in the other order, with the sign of the Selling parameters reversed; no reflection is applied. Which zeros a pattern has also fixes the kind of face: for the three-zero sorts C5, T5 and O5 the zeros are ab, ac and bc, a triangle on a, b, c, and the same lattices also have reduced cells whose zeros form a path.*

sort	S^6 order	this paper	sort	S^6 order	this paper
C1	rrrrrr	rrrrrr	O2	rs0srt	srt0sr
C3	rr0rr0	rr00rr	O3	rs0rs0	rs00sr
C5	000rrr	rrr000	O4	00rsst	sstr00
H4	00rrrs	rrsr00	O5	000rst	rst000
T1	rrsrrs	rrssrr	M1A	rrsttu	ttusrr
T2	rr0rrs	rrs0rr	M1B	rstrsu	rsutsr
T5	000rrs	rrs000	M2A	rs0stu	stu0sr
R1	rrrsss	sssrrr	M2B	rs0rst	rst0sr
R3	rr0sr0	sr00rr	M3	rs0ts0	ts00sr
O1A	rrsrrt	rrtsrr	M4	00rstu	stur00
O1B	rstrst	rsttsr	A1	rstuvw	uvwtsr
			A2	rs0tuv	tuv0sr
			A3	rs0tu0	tu00sr

Table S8. *Pseudo-symmetric PDB entries (Le Page lattice above the reported type at 3°) by the Delone sort of the Le Page lattice, operators of the reported lattice's own holohedry excluded, entries with an exact higher lattice included (their smallest price is 0): number of entries, and the 10th/50th/90th percentiles of the largest obliquity δ (degrees) and of the largest and smallest price (%) among the pseudo-symmetry operators of each entry. Sorts with no entries are omitted. The transitions from the reported type are in Table S9. From* ***results_pdb_lepage.txt****.*

sort	type	entries	δ (°)	largest price (%)	smallest price (%)
C1	cI	593	0.14/1.64/2.73	0.69/8.10/13.58	0.32/7.13/13.18
C3	cF	562	0.19/1.72/2.76	0.95/8.51/13.61	0.50/7.68/13.41
C5	cP	1 349	0.29/1.70/2.72	1.43/8.40/13.45	0.47/5.52/12.59
H4	hP	2 008	0.08/1.40/2.70	0.40/6.91/13.32	0.11/4.03/12.70
T1	tI	1 376	0.06/1.32/2.67	0.28/6.60/13.33	0.17/5.55/12.60
T2	tI	330	0.05/1.48/2.69	0.24/7.30/13.43	0.11/4.98/12.44
T5	tP	14 816	0.22/1.47/2.69	1.07/7.27/13.43	0.70/6.56/12.98
R1	hR	96	0.04/1.55/2.76	0.22/7.64/13.78	0.13/2.98/13.13
R3	hR	393	0.07/1.82/2.81	0.37/9.09/13.86	0.29/7.64/13.72
O1A	oF	1 126	0.23/1.51/2.70	1.12/7.45/13.33	1.02/7.05/13.17
O1B	oI	331	0.10/1.58/2.78	0.49/7.80/13.73	0.36/7.63/13.73
O2	oI	1 078	0.14/1.40/2.73	0.70/6.92/13.50	0.48/6.33/13.36
O4	oS	11 679	0.14/1.49/2.74	0.67/7.36/13.56	0.62/7.15/13.51
O5	oP	4 760	0.03/1.08/2.59	0.17/5.33/12.79	0.15/5.13/12.69
M1A	mC	867	0.72/2.09/2.83	3.53/10.33/13.98	3.53/10.33/13.98
M1B	mC	398	0.12/1.80/2.72	0.57/8.91/13.41	0.57/8.91/13.41
M2A	mC	854	0.13/1.75/2.77	0.66/8.64/13.66	0.66/8.64/13.66
M2B	mC	2	0.61/0.95/1.29	3.02/4.70/6.38	3.02/4.70/6.38
M3	mC	1	2.39/2.39/2.39	11.80/11.80/11.80	11.80/11.80/11.80
M4	mP	582	0.10/1.40/2.65	0.47/6.89/13.07	0.47/6.89/13.07

Table S9. *The cheapest pseudo-symmetric alternative of PDB entries without an exact higher lattice (operators of the reported lattice's own holohedry excluded), by transition from the reported Bravais type to the one Le Page's* 3° *finds: entries with priced alternatives, the median price (%), and the share within* 1%*, within the 90th percentile of non-isomorphism (*4.2%*) and within its 95th (*6.7%*). The sixteen most frequent transitions.*

transition	entries	median	≤ 1%	≤ 4.2%	≤ 6.7%
oP→tP	12 120	6.98	9	31	48
mP→oS	9 474	7.38	12	33	46
mP→oP	4 417	5.78	20	42	55
aP→mC	2 113	9.49	9	19	30
mC→oS	1 831	7.41	12	33	45
mP→tP	1 542	3.66	27	54	69
mC→oI	1 308	7.13	11	30	48
oS→hP	1 169	7.34	10	26	45
oI→tI	1 165	6.25	19	38	52
mC→oF	1 067	7.29	8	29	46
oS→tP	880	7.42	14	32	46
tP→cP	724	7.10	7	28	47
aP→mP	571	6.97	15	30	48
mP→hP	479	0.71	61	91	100
mC→hR	391	8.92	19	29	37
mC→tI	391	4.56	23	46	64
all	42 656	6.69	14	35	50

Over the sorts with more than two entries, the median largest price lies between 5.3% (O5, oP) and 10.3% (M1A); the median largest obliquity between 1.1° and 2.1°. Within a sort the median largest price of the operators of order 3, 4 or 6 is within half a percentage point of that of the twofolds (`results_pdb_lepage.txt`).

The closures of the 24 sorts. For each face the arrows are the operators of $\mathcal{U}$ that map the face into the cone; on the face every arrow permutes the six conorm slots, so each fixes a subspace of conorm vectors, and the sorts are the closed sets of equalities obtained by intersecting those subspaces (Definition A5 of the main text). For a sort, the point group is the set of arrows that fix the sort's subspace pointwise, the visible symmetries are those that are swaps of the reduced cell, and two superbases have the same shape when their conorms agree up to a slot permutation induced by a swap. Table S10 is the resulting catalogue; every entry is the result of this linear algebra on integer matrices. In every sort the number of symmetries out of each labelled superbase is the order of the holohedry, and the twofolds, with the inversion, generate the group.

Symmetry visible in one cell. In the sorts with zero conorms the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal. None of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic. The orbit of the reduced cell under the point group has |group|/visible superbases, all of its shape: for oP the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal. Reduction of a measured orthorhombic cell stops at whichever of the sixteen superbases its errors favour; an orthogonal superbase is always one flip away, but the one in the reference's labelling can be four flips away, and no twofold of the lattice reaches it. The catalogues of shapes and of the conditions under which one sort becomes another are Tables S1 and S2.

Table S10. *The Selling closure of each Delone sort, computed from the operators alone. Patterns are conorms in the slot order* da, db, dc, ab, ac, bc*; equal letters denote equal values and* 0 *a zero; they are the patterns of Andrews & Bernstein (2023) read in this order (Table S7). Superbases are counted up to swaps. "Visible" is the number of symmetries that relabel the reduced superbase written in the pattern shown; another superbase of the same lattice can show a different number. "Shapes" is the number of distinct cell shapes among the superbases, equal in every case to the number of orbits of superbases under the symmetries.*

sort	Bravais	pattern	zeros	superbases	|group|	visible	shapes
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr00rr	2	3	48	16	1
C5	cP	rrr000	3	16	48	12	2
H4	hP	rrsr00	2	6	24	4	1
T1	tI	rrssrr	0	1	16	16	1
T2	tI	rrs0rr	1	2	16	8	1
T5	tP	rrs000	3	16	16	4	3
R1	hR	sssrrr	0	1	12	12	1
R3	hR	sr00rr	2	3	12	4	1
O1A	oF	rrtsrr	0	1	8	8	1
O1B	oI	rsttsr	0	1	8	8	1
O2	oI	srt0sr	1	2	8	4	1
O3	oI	rs00sr	2	3	8	8	2
O4	oS	sstr00	2	6	8	4	2
O5	oP	rst000	3	16	8	2	4
M1A	mC	ttusrr	0	1	4	4	1
M1B	mC	rsutsr	0	1	4	4	1
M2A	mC	stu0sr	1	2	4	2	1
M2B	mC	rst0sr	1	2	4	4	2
M3	mC	ts00sr	2	3	4	4	2
M4	mP	stur00	2	6	4	2	3
A1	aP	uvwtsr	0	1	2	2	1
A2	aP	tuv0sr	1	2	2	2	2
A3	aP	tu00sr	2	3	2	2	3

S6. Distances to the 24 sorts

The same computation gives the distance from a measured cell to each of the 24 sorts. In each of the 6096 frames of $\mathcal{U}$ the cell's conorms are projected, in the strain, onto the sort's subspace (its zeros and equalities, Definition A5 of the main text, the other conorms non-negative), and the smallest distance is taken. This is the computation of SELLA (Andrews *et al.*, 2023), which projects the boundary images of a cell in S^6 onto the same subspaces, and of Oishi-Tomiyasu (2012) in Selling coordinates; projection onto Bravais-lattice subspaces goes back to Andrews & Bernstein (1988). What

changes is the norm, which is the same in every frame, and the set of frames, which is $\mathcal{U}$. We call the 24 distances the *sort spectrum* of the cell. A distance below the adjacency radius r_1 is exact, since the frames of $\mathcal{U}$ then contain the frame of the nearest lattice of the sort; a minimum above r_1 is an upper bound over the frames used, the true distance being at least r_1, and the implementation flags it (`certified_sort_spectrum`). Along every specialisation of Table S2 the distance cannot decrease, since the lower sort's subspace lies inside the upper one's.

Fig. S1 draws the spectrum of the worked cell on the graph of specialisations. The nearest sorts are A2 at 0.28%, M4 (mP) at 0.78%, A3 at 3.8% and O5 (oP) at 5.1%, and every other sort is at least 25% away, a bound that holds because r_1 reaches the cap for this cell. For a tetragonal lysozyme cell indexed in $P1$ the twelve sorts from aP to tP all lie within 0.22% (Fig. S2).

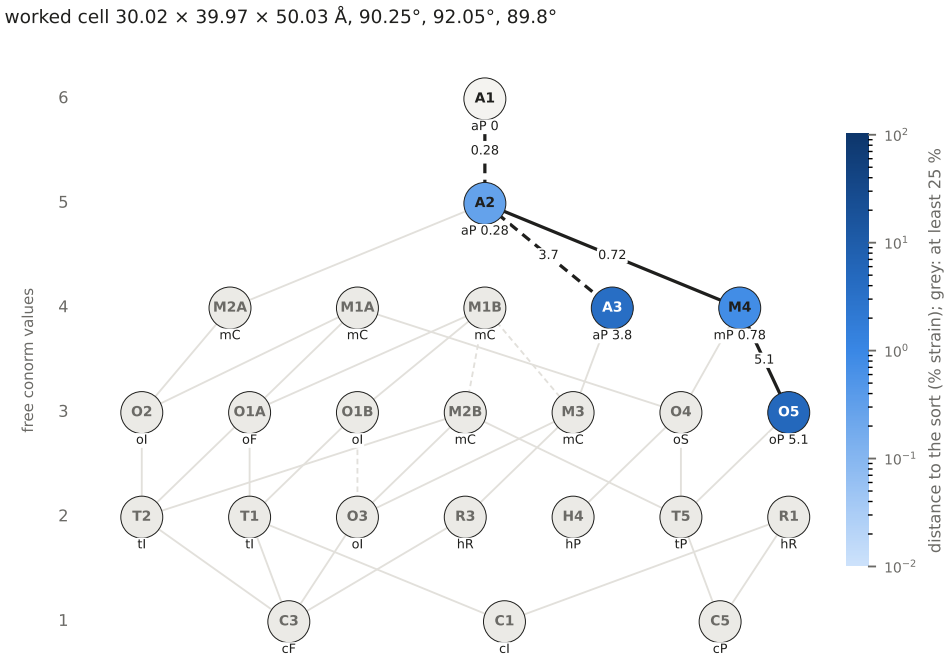


Fig. S1. The 24 Delone sorts as a graph for the measured worked cell. Rows are the number of free conorm values. Each node gives the Bravais type and the strain distance (%) from the cell to the sort, its colour the same distance on a log scale; sorts at least 25% away are grey. Edges are the specialisations of Table S2, labelled with the distance from the cell idealised onto the upper sort to the lower one. Dark edges are routes on which these steps add in quadrature to the lower sort's distance within 3%, drawn where that distance is below 10%. Dashed edges keep the Bravais type.

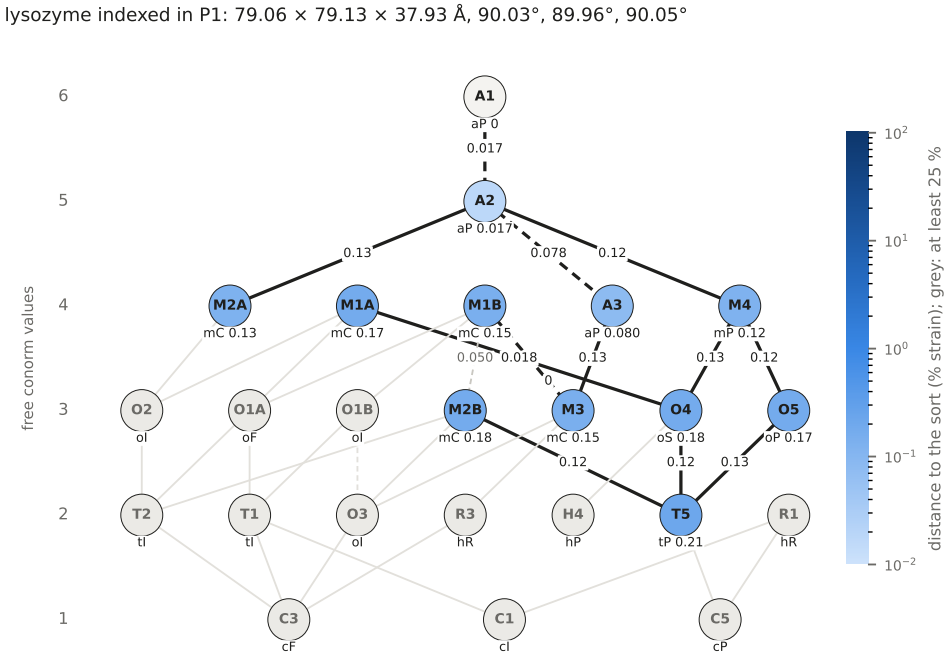


Fig. S2. The 24 Delone sorts as a graph for a tetragonal lysozyme cell (79.1 × 79.1 × 37.9 Å) indexed in $P1$ with an added error, drawn as Fig. S1. The twelve sorts from aP to tP lie within 0.22%; every other sort is at least 25% away (grey).

Le Page's 3° *by sort.* The distance to a sort at Le Page's boundary depends on the sort, while the price of each operator does not. For exact lattices of each sort perturbed in random directions until Le Page's obliquity for the sort's Bravais group is exactly 3°, the median largest twofold price is 14.8% in every sort, which Proposition A9 makes true by construction and the table confirms numerically (larger only in the monoclinic tails, up to 40–55%, where Le Page's group is another C-centred group), while the median distance to the sort is 7.4–7.7% for the orthorhombic and monoclinic sorts other than O3, M2B and M3, 7.9–8.1% for tP, tI, hP and hR, and 8.3–8.5% for the cubic sorts, with longer tails for the boundary sorts O3, M2B and M3 (Table S11). A single band on the distance to a sort therefore tests each sort at a different level, which is why the reindexing in this paper is decided on the prices of operators.

Table S11. *Le Page's* 3° *by Delone sort: for exact lattices of each sort, each perturbed in a random direction until the obliquity of the sort's Bravais group is* 3°*, the strain distance to the sort (median, and 10th–90th percentiles) and the median of the largest price of the lattice's twofolds and of its other rotations, by order, in %. In the monoclinic sorts some perturbations reach the boundary through a different C-centred group, whose own twofold costs* 14.8% *while the exact lattice's costs more. Of 25 directions per sort, those that never reach an obliquity of* 3° *are dropped, hence* $n < 25$ *in seven sorts; with* $n \leq 25$ *the percentiles are indicative.*

sort	type	n	distance to sort	twofold price	other orders
C1	cI	25	8.3 (7.6–9.0)	14.8	3: 14.1, 4: 15.0
C3	cF	25	8.3 (7.7–9.2)	14.8	3: 14.3, 4: 14.5
C5	cP	25	8.5 (7.6–9.3)	14.8	3: 14.2, 4: 14.7
H4	hP	25	7.9 (7.5–8.5)	14.8	3: 13.7, 6: 11.3
T1	tI	25	7.9 (7.5–9.1)	14.8	4: 14.0
T2	tI	25	8.0 (7.6–8.8)	14.8	4: 14.4
T5	tP	25	8.0 (7.4–8.9)	14.8	4: 14.1
R1	hR	25	8.0 (7.5–8.6)	14.8	3: 14.4
R3	hR	25	8.1 (7.5–8.8)	14.8	3: 14.5
O1A	oF	24	7.5 (7.4–7.8)	14.8	—
O1B	oI	25	7.6 (7.4–8.1)	14.8	—
O2	oI	25	7.6 (7.4–8.2)	14.8	—
O3	oI	25	8.6 (7.6–13.6)	14.8	—
O4	oS	25	7.7 (7.4–8.2)	14.8	—
O5	oP	25	7.5 (7.4–8.3)	14.8	—
M1A	mC	20	7.4 (7.4–9.1)	14.8	—
M1B	mC	15	7.4 (7.4–7.4)	14.8	—
M2A	mC	18	7.4 (7.4–8.5)	14.8	—
M2B	mC	17	9.3 (7.5–19.9)	14.8	—
M3	mC	23	8.9 (7.5–16.1)	14.8	—
M4	mP	23	7.4 (7.4–7.4)	14.8	—

Against BLDConograph and SELLA. On the seven sample cells that Oishi-Tomiyasu (2024) distributes with BLDConograph, the highest Bravais type within 7.4% is the program's type in all seven, with the next type at least 14.9% away (Table S5). SELLA (`CmdSella`, Andrews *et al.*, 2023, default settings) was run on the worked cell, the lysozyme cell, a pseudo-tetragonal cell, a cell with all angles 92°, a needle, two plates and the seven BLDConograph cells. The two rankings of the 24 sorts agree closely, with Spearman's rank correlation 0.77 to 0.97 over all 24 sorts, which the many distant sorts inflate. The two scales are not proportional: SELLA's distance divided by the cell's mean squared edge is a median 0.71 times the strain distance, with a 10th to 90th percentile range of 0.05 to 26 (`results_sella.txt`). With SELLA's band set

by that median ratio, which is calibrated on the same 14 cells and so flatters the agreement, the Bravais types within the band agree in 11 of the 14 cells. They differ on the needle and the two plates, where the normalisation by the mean squared edge, our choice and not SELLA's, weighs the long edges differently: SELLA admits hP, hR, oF, tI and tP for the needle, which are far in strain, and the strain band admits oF (and tI) for the plates, which SELLA's does not. For these thin cells r_1 is 4 to 6%, so their strain distances above it are bounds and are flagged.

S7. Computational checks

Each entry of Table S12 names the test files that reproduce the numbers of a section, table or figure of the main text. A mark z*nn* is `manuscript/appendix_z/test_z`*nn*`_*.py` in the `agentsg` repository, lib is `agentsg/tests/test_strain_closure.py`, and bench names a script in `manuscript/benchmark`. The checks were run with Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1 and cctbx-base 2025.11, and the comparisons with cctbx, SELLA (CmdSella) and the NCDist/CS6Dist libraries; the checks z24, z25 and z42 are skipped when their external data are absent, and none of the numbers of the paper rests on them; the PDB scripts need the Zenodo snapshot (doi:10.5281/zenodo.22986222) in `data/`.

Formal proofs. The Lean development in `manuscript/formal` builds with Lean 4.35.0 and Mathlib release v4.35.0 (`lake build`). Its data file is generated from `universal_operators_columns.json` and records the file's SHA-256 hash, which `check_data.py` recomputes. `audit/axioms.lean` prints the axioms of each headline theorem. Beyond the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`), the completeness of $\mathcal{U}$ (`arrow_mem_U`) depends on one compiled evaluation (`native_decide`, of `finite_check`); its converse (`U_arrow`) on two (`U_realised`, `U_unimodular`); the dependence on the zero pattern (`arrow_zero_pattern`) on two (`finite_check`, `zero_`

`pattern_check`); and the closure of $\mathcal{U}$ under inversion and under the 48 swaps on one each. The window (`box_of_conn`, Lemma A2 of the main text), Lemma A1 (`excess_identity`), the shortest partial sums (`partial_shortest`), every part of Propositions A10 and A9(i) (`price_mul`, `price_inv`, `coset_lower`, `coset_upper`, `Hgroup`, `candidates_eq_coset`, `stab_sub_H`, `H_sub_H`, `price_le_two_dist_fixed`, `twofold_midpoint_fixed`) and Proposition A11(i)–(ii) depend on the standard axioms alone, that is, on the kernel.

Certified radii. The depth of a superbase F with conorms q under the metric is the least strain norm $\|r\|_M$ of a change $r \geq -q$ of its conorms, with M the strain norm in the frame of F ($r^{\mathsf{T}} M r = \operatorname{tr}(\mathbf{G}^{-1}\Delta\mathbf{G}\,\mathbf{G}^{-1}\Delta\mathbf{G})$). For any multipliers $\lambda \geq 0$ with $c = -\lambda \cdot q > 0$ the depth is at least $c/(\lambda^{\mathsf{T}} M^{-1} \lambda)^{1/2}$, by weak duality, and any r with $q + r \geq 0$ whose metric is positive definite bounds it from above. `certify_radii.py` evaluates the balls of Lemma A6 and the conorm test of Proposition A4 exactly, so that every superbase within the cap is among its candidates, proves a lower bound for every candidate outside the layer and an upper bound for the nearest, all in rational arithmetic; floating point only proposes λ and r. The certificates are for the reduced conorms as stored in double precision, read as exact rationals, which agree with the cells of Table 8 to about 15 digits.

Timings. On a tenth of the benchmark, widening a reference took a median of 0.02 s and reindexing one new cell 0.03 s (90th percentiles 0.03 and 0.12 s); a $150 \times 120 \times 4.8$ Å plate took 0.9 s to widen. A $300 \times 300 \times 3$ Å plate takes about a minute to widen and its r_1 several minutes, and thinner cells are refused by the cap on the ball. A vectorised lookup of $\mathcal{U}$ for one new cell, reductions included, takes about 0.04 s (`adjacency_radii.py`).

Table S12. *The checks behind each part of the main text.*

main text	checks
§2.1 faces; §2.2 Table 1	z52, z56 (the 38 faces); z46 (flip matrices)
§3.1 counts of superbases and objects	z54, z56
§3.2, Table 3, Fig. 1	z46 (the ring, the action of $2/m$, the word *ab.bc.db*)
§3.3, Table 4	z52 (6096, inversion, window, nine kinds), z56 (every window member in $\mathcal{U}$, entries by zero pattern; about 25 min); bench: `face_labellings.py`, `export_universal_operators.py`
§3.4, Table 5	z59 (balls, layers of 127 and 2755), z60 (twofold price from obliquity, Lemma A7, Proposition A10); bench: `lookup_plate_case.py`
§3.5, Table 6	z52, z56 ($[-2,2]$ labelling), z58, z59 (brute force)
§4.1–4.2, Table 7	z59 (filtered layer), z44, z45 (difference-wall members, walks); lib (reindexing, scale bound, volume check, centring and input checks, the column form of $\mathcal{U}$, classes)
§4.3–4.4	bench: `make_benchmark.py`, `window_failures.py`, `pdb_nonisomorphism.py`, `pdb_band_controls.py`
§5.1	z60 (Proposition A11 near oP); bench: `niggli_types.py`, `niggli_presentation.py`
§5.2, Fig. 2, Table 8	z60 (depths, growth of the set), z57, z58, z59; bench: `adjacency_radii.py`, `certify_radii.py`, `evaluate_strain.py`, `lookup_benchmark.py`, `window_failures.py`, `niggli_window.py`
§5.3, Table 9	lib, z60 (classes, prices, centred references); bench: `coset_rank.py`
§5.4, Fig. 3, Table 10, Tables S8, S9	bench: `pdb_nonisomorphism.py`, `pdb_band_controls.py`, `pdb_lepage.py`, `pdb_lepage_null.py`, `pdb_pricing_summary.py`, `pdb_band_uncertainty.py`, `pdb_band_coverage.py` (need the Zenodo snapshot)
Appendix A	z61 (the terms: obtuse superbases and shortest partial sums, the flip at a zero conorm), z44 (A1), z56 (A2), z61 (A3, A4), z53 (A5), z59 (A6, A8), z60 (A7, A9, A10(i)–(ii), A11(iii)), z61 (A10(iii)–(v) by brute-force enumeration, A11(i)); `manuscript/formal` (Lean)
§S6, Figs S1, S2, Tables S11, S5	z58, z60, z51; bench: `lepage_calibration.py`, `compare_sella.py`
§S4	z36, z44 (Lemma S4.4), z45, z46, z47, z55, z57
§S5, Table S10	z53, z54, z46, z49 (Table S1), z50 (Table S2)
§S8, Table S13	bench: `compare_methods.py`, `benchmark_misses.py`, `window_failures.py`

S8. Existing searches on the same cases

The test cases of the main text were also given to the searches that data-processing programs use, each priced in the same way and kept within the same band: every matrix with entries in $\{-1,0,1\}$ or $\{-2,\dots,2\}$ between the two Selling-reduced cells; the $\{-1,0,1\}$ search between the two Niggli cells; cctbx `similarity_transformations`

between the two Niggli cells (Grosse-Kunstleve *et al.*, 2002) with generator ranges 1 and 2; and a pipeline that runs the cctbx lattice-symmetry analysis (Sauter *et al.*, 2004) on both cells and returns every operation of the reference's highest lattice group between the two standard settings. Table S13 gives the counts. The searches between Niggli cells and the cctbx similarity searches found the true operator in every case; the fixed windows between Selling-reduced cells, which no program uses, miss operators where Table 4 of the main text says they can; the pipeline missed a needle whose reference and measured cell idealise at 3° to different lattices, and a few stress and window cases. The purpose of the table is to show that the guarantee and the existing searches agree where both apply, not to rank them.

Table S13. *Existing searches on the test cases of the main text (Section 4.4): the true operator among the candidates, by method and family; candidates priced by the geodesic distance after the best scale and kept within a band of* 7.4% *(the reference widened at* 7.7% *for the closure and the lookup). Window family: cells measured in a member that needs an entry of* 3*, and near-mP and near-orthogonal lattices measured twice with an error of* 0.1 *to* 2%*, the reference given as its Niggli cell (split by Niggli type) or in a random basis (split by the kind of face). A dash: not run. The lookup on the benchmark and stress families is in* `results_lookup.txt`*. Results in* `manuscript/benchmark/`*.*

					2- or 3-zero faces, reference given as			
		stress			Niggli cell		random basis	
method	benchmark	0.5ϵ	0.9ϵ	entry 3	type I	type II	2 zeros	3 zeros
closure	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
lookup of $\mathcal{U}$	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
$\{-1, 0, 1\}$, Selling cells	778/780	58/60	52/60	24/60	142/171	129/129	120/150	99/150
$\{-2, \ldots, 2\}$, Selling cells	780/780	59/60	60/60	52/60	170/171	129/129	150/150	149/150
$\{-1, 0, 1\}$, Niggli cells	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
cctbx similarity, range 1	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
cctbx similarity, range 2	780/780	60/60	60/60	60/60	171/171	129/129	150/150	150/150
lattice-symmetry pipeline	779/780	56/60	54/60	60/60	171/171	126/129	147/150	148/150

S9. Uncertainty of the PDB statistics, chance, and the Niggli window

Intervals. Table S14 gives block-bootstrap 95% intervals for the percentiles of Table 10 of the main text: the deposition blocks (runs of id-adjacent pairs) are resampled with replacement, 1000 times, which keeps the correlation within a deposition series. The

shares of entries with an alternative within the band have intervals of ±0.2 to ±0.4 points, where a type with fewer than 200 blocks uses the pooled band (`pdb_band_uncertainty.py`).

Table S14. *Non-isomorphism in the PDB by reported Bravais type with block-bootstrap 95% intervals (cut* 20%*, identity as the operator, price after the scale fit, %).* [a] *Fewer than 200 deposition blocks; the pooled band stands in for these types in the shares of the main text, and the cubic rows are zero by construction.*

type	pairs	blocks	median	90th	95th
aP	902	535	2.06 (1.82–2.27)	7.66 (6.40–9.46)	11.68 (9.76–13.70)
mP	7 077	3 277	1.27 (1.21–1.33)	5.38 (5.01–5.84)	9.02 (8.21–9.78)
mC	4 932	1 977	1.49 (1.41–1.59)	5.06 (4.73–5.49)	8.40 (7.16–9.23)
oP	13 615	5 987	1.11 (1.07–1.14)	4.70 (4.46–4.95)	7.43 (7.06–7.81)
oS	2 271	1 108	1.01 (0.94–1.08)	3.57 (3.32–4.01)	5.50 (4.96–6.14)
oI	1 353	574	1.22 (1.07–1.39)	5.12 (4.28–6.16)	7.91 (6.40–9.15)
oF[a]	31	22	1.99 (1.23–3.10)	5.87 (3.39–8.28)	7.54 (4.48–8.95)
tP	5 174	2 061	0.52 (0.48–0.57)	2.59 (2.36–2.85)	4.00 (3.61–4.41)
tI	805	416	0.65 (0.53–0.78)	2.76 (2.28–3.34)	4.91 (3.61–6.03)
hR	1 270	597	0.68 (0.61–0.79)	3.54 (3.10–3.94)	5.86 (4.59–6.71)
hP	6 817	3 167	0.57 (0.53–0.62)	2.87 (2.65–3.04)	4.56 (4.22–4.95)
cP[a]	363	178	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
cI[a]	385	194	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
cF[a]	211	106	0.00 (0.00–0.00)	0.00 (0.00–0.00)	0.00 (0.00–0.00)
all	45 222	20 206	0.98 (0.96–1.00)	4.21 (4.09–4.33)	6.72 (6.52–6.98)

Against chance. Shuffling the free cell parameters independently among the entries of one space group keeps their distributions and destroys any relation between them; shuffling within twenty classes of cell volume keeps cells of one size together. At Le Page's 3° the real rate exceeds both shuffled ones for $P1$ (42–45% against 30–41%) and $P2_12_12_1$ (23–24% against 13–17%), whose real cells are more often pseudo-tetragonal, exceeds them slightly for $P2_1$ (46% against 39–44%), and matches them for $C2$ and $P2_12_12$. Near-exact pseudo-symmetry is not chance: the share whose cheapest alternative costs at most 1% is 10–11% for real $P1$ cells against 2–4% shuffled, and 9% against 5–6% for $P2_1$ (intervals about ±1 point). At 4.2% the excess remains for $P1$ (16% against 6–10%) but not clearly for $P2_1$ (18% against 13–18%, intervals about ±2 points), and mP entries carry much of the share within 4.2% (two draws of 1500 cells per arm; `results_pdb_lepage_null.txt`).

The crystal-form proxy. Random pairs of entries with one space group whose identifiers do not share their first three characters, drawn with the space-group mix of

the id-adjacent pairs, pass the 20% cut in 1.5% of 19 986 cases (0.75% at 5%; `pdb_proxy_check.py`). Unrelated crystals rarely pass the cut by chance, so the movement of the percentiles with the cut reflects variation within deposition series rather than unrelated pairs.

The Niggli window. `niggli_window.py` draws 100 lattices on each of the 38 faces (half of them anisotropic, with vertex scales up to 30) and on each of the 24 sorts, measures each twice with independent errors of total strain 0.5, 2, 5 and 7.7%, the new cell in a random basis, and computes the true operator between the two Niggli cells (cctbx) and between the two Selling-reduced cells directly. Between Niggli cells its entries were in $\{-1, 0, 1\}$ in all 24 800 cases, strictly and not only up to the lattice's point group, and in 3 040 further cases on strongly anisotropic lattices (`niggli_window.py thin`: 40 per face, vertex scales up to 60, the square root of the metric's eigenvalue ratio up to 98, median 5); between Selling-reduced cells an entry of 2 or 3 was needed in 3 765 cases (on two zeros sharing a vector, three zeros, and the sorts cP, tP and oP) and in none once composed with the point group.

References

Andrews, L. C. & Bernstein, H. J. (1988). *Acta Cryst.* A**44**, 1009–1018.

Andrews, L. C. & Bernstein, H. J. (2023). *Acta Cryst.* A**79**, 499–503.

Andrews, L. C., Bernstein, H. J. & Pelletier, G. A. (1980). *Acta Cryst.* A**36**, 248–252.

Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2023). SELLA – a program for determining Bravais lattice types. arXiv:2303.03122.

Conway, J. H. & Sloane, N. J. A. (1992). *Proc. R. Soc. London Ser. A*, **436**, 55–68.

Delaunay, B. (1933). *Z. Kristallogr.* **84**, 109–149.

Grosse-Kunstleve, R. W., Sauter, N. K., Moriarty, N. W. & Adams, P. D. (2002). *J. Appl. Cryst.* **35**, 126–136.

Hahn, Th. (2006). Editor. *International Tables for Crystallography*, Vol. A, 5th ed. Dordrecht: Springer.

Oishi-Tomiyasu, R. (2012). *Acta Cryst.* A**68**, 525–535.

Oishi-Tomiyasu, R. (2024). *Acta Cryst.* A**80**, 339–350.

Sauter, N. K., Grosse-Kunstleve, R. W. & Adams, P. D. (2004). *J. Appl. Cryst.* **37**, 399–409.

Selling, E. (1874). *J. Reine Angew. Math.* **77**, 143–229.

Zwart, P. H. (2026). `agentsg`: a space-group and lattice engine, version 0.5.0. https://github.com/phzwart/agentsg.